Build Variants Documentation

Overview

Rhythm uses Android build variants (product flavors) to create different builds for different distribution channels while maintaining a single codebase.

Quick Reference

```
# Build commands
./gradlew assembleFdroidRelease      # For F-Droid
./gradlew assembleGithubRelease      # For GitHub Releases

# List all build variants
./gradlew tasks --all | grep assemble
```

Build Configuration

Flavor Dimension: `distribution`

All flavors belong to the `distribution` dimension, ensuring they're mutually exclusive.

Product Flavors

1. **`fdroid`**

Purpose: F-Droid repository distribution
Package: `chromahub.rhythm.app`
Version Suffix: `-fdroid`

BuildConfig Flags:

```
ENABLE_YOUTUBE_MUSIC = true
ENABLE_SPOTIFY_CANVAS = true
ENABLE_APPLE_MUSIC = true
ENABLE_DEEZER = true
ENABLE_LRCLIB = true
ENABLE_SPOTIFY_SEARCH = true
```

Default Settings: APIs and auto-updates disabled by default (opt-in for F-Droid compliance)

Philosophy: FOSS users expect maximum functionality. F-Droid isn't bound by corporate app store policies, but opt-in defaults ensure compliance.

- `github`

Purpose: Direct GitHub releases
Package: `chromahub.rhythm.app`
Version Suffix: `-gh`

BuildConfig Flags:

```
ENABLE_YOUTUBE_MUSIC = true
ENABLE_SPOTIFY_CANVAS = true
ENABLE_APPLE_MUSIC = true
ENABLE_DEEZER = true
ENABLE_LRCLIB = true
ENABLE_SPOTIFY_SEARCH = true
```

Default Settings: APIs and auto-updates enabled by default (opt-out available)

Use Case: Users who download directly from GitHub releases get the full-featured experience.

How BuildConfig Flags Work

. Compile-Time Conditional Compilation

Flags are defined in `app/build.gradle.kts`:

```
buildConfigField("boolean", "ENABLE_YOUTUBE_MUSIC", "true")
```

. Runtime API Initialization

APIs are initialized conditionally in `NetworkClient.kt`:

```
val ytmusicApiService: YTMusicApiService? = if (BuildConfig.ENABLE_YOUTUBE_MUSIC) {
    ytmusicRetrofit.create(YTMusicApiService::class.java)
} else null
```

. Settings UI Adaptation

UI components check BuildConfig before rendering:

```
if (BuildConfig.ENABLE_YOUTUBE_MUSIC) {
    ApiServiceRow(title = "YouTube Music", ...)
}
```

. Runtime Feature Checks

Code checks both BuildConfig AND user settings:

```
fun isYTMusicApiEnabled(): Boolean =
    BuildConfig.ENABLE_YOUTUBE_MUSIC && (appSettings?.ytMusicApiEnabled?.value ?: false)
```

Default Settings by Variant:

- **F-Droid:** APIs and auto-updates disabled by default (opt-in for compliance)
- **GitHub:** APIs and auto-updates enabled by default (opt-out available)

This ensures:

- Build variants properly control feature availability
- Users can toggle features on/off within allowed limits
- F-Droid compliance through opt-in defaults
- Single codebase maintains all variants

Variant-Specific Resources

Directory Structure

```
app/src/
├── main/                    # Shared code and resources
├── fdroid/
│   └── res/
│       └── values/
│           └── config.xml
└── github/
    └── res/
        └── values/
            └── config.xml
```

Resource Flags

Each variant has a `config.xml` with boolean resources:

fdroid/res/values/config.xml:

```
<bool name="feature_youtube_music_available">true</bool>
<bool name="feature_spotify_canvas_available">true</bool>
<string name="distribution_channel">F-Droid</string>
```

github/res/values/config.xml:

```
<bool name="feature_youtube_music_available">true</bool>
<bool name="feature_spotify_canvas_available">true</bool>
<string name="distribution_channel">GitHub</string>
```

Build Output

APK Naming Convention

```
Rhythm-{versionName}-{suffix}-{flavor}-{buildType}.apk
```

Examples:

```
Rhythm-4.0.312.858-fdroid-fdroid-release.apk
Rhythm-4.0.312.858-gh-github-release.apk
```

Output Locations

APKs:

```
app/build/outputs/apk/fdroid/release/
app/build/outputs/apk/github/release/
```

App Bundles (AAB):

```
app/build/outputs/bundle/fdroidRelease/app-fdroid-release.aab
app/build/outputs/bundle/githubRelease/app-github-release.aab
```

Testing Variants

Build All Variants

```
# Release builds
./gradlew assembleRelease

# Debug builds
./gradlew assembleDebug

# Specific variant
./gradlew assembleFdroidDebug
```

Install Specific Variant

```
# Install F-Droid debug
./gradlew installFdroidDebug

# Install GitHub release
adb install app/build/outputs/apk/github/release/Rhythm-*.apk
```

Verify BuildConfig Values

After building, check generated BuildConfig:

F-Droid:

```
app/build/generated/source/buildConfig/fdroid/release/chromahub/rhythm/
app/BuildConfig.java
```

Should contain:

```
public static final boolean ENABLE_YOUTUBE_MUSIC = true;
public static final boolean ENABLE_SPOTIFY_CANVAS = true;
```

GitHub:

```
app/build/generated/source/buildConfig/github/release/chromahub/rhythm/
app/BuildConfig.java
```

Should contain:

```
public static final boolean ENABLE_YOUTUBE_MUSIC = true;
public static final boolean ENABLE_SPOTIFY_CANVAS = true;
```

CI/CD Integration

GitHub Actions Example

```yaml
name: Build All Variants

on:
  push:
    branches: [ main ]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Set up JDK 17
        uses: actions/setup-java@v3
        with:
          java-version: '17'

      - name: Build F-Droid variant
        run: ./gradlew assembleFdroidRelease

      - name: Build GitHub variant
        run: ./gradlew assembleGithubRelease

      - name: Upload artifacts
        uses: actions/upload-artifact@v3
        with:
          name: apk-variants
          path: app/build/outputs/apk/**/release/*.apk
```

F-Droid Metadata

Add to `.fdroid.yml` or `metadata/*.yml`:

```
Builds:
  - versionName: 4.0.312.858
    versionCode: 40312858
    gradle:
      - fdroid

    # F-Droid will automatically use the fdroid flavor
```

Troubleshooting

Issue: Wrong variant built

Symptom: APK has unexpected features enabled/disabled
Solution: Verify you ran the correct assemble task (e.g., `./gradlew assembleFdroidRelease`)

Issue: UI shows disabled features

Symptom: F-Droid build hides YouTube Music settings
Solution: Check `BuildConfig.ENABLE_YOUTUBE_MUSIC` in generated files

Issue: Null pointer when calling API

Symptom: Crash when accessing `ytmusicApiService`
Solution: Ensure null-safe access:

```
if (ytmusicApiService != null) {
    ytmusicApiService.search(...)
}
```

Issue: Build fails with variant not found

Symptom: `Task 'assembleFdroid' not found`

Solution: Use full variant name: `assembleFdroidRelease` or `assembleFdroidDebug`

Maintenance

Adding a New Build Variant

1. **Add flavor in `app/build.gradle.kts`:**

```
create("newvariant") {
    dimension = "distribution"
    applicationId = "chromahub.rhythm.app"
    buildConfigField("boolean", "ENABLE_YOUTUBE_MUSIC", "true")
    versionNameSuffix = "-new"
}
```

1. **Create variant directory:**

```
mkdir -p app/src/newvariant/res/values
```

1. **Add config.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="distribution_channel">New Variant</string>
</resources>
```

1. **Test build:**

```
./gradlew assembleNewvariantRelease
```

Removing a Build Variant

1. Delete flavor block from `build.gradle.kts`
2. Remove `app/src/variantname/` directory
3. Sync Gradle: `./gradlew --refresh-dependencies`

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[1]

Best Practices

. Version Consistency

[2]

- All variants share the same `versionCode` and `versionName`
- Only `versionNameSuffix` differs for logging

[3] . Feature Parity

- F-Droid and GitHub should always have identical features

. Testing

- Test each variant separately before release

[4]

- Ensure disabled features don't appear in UI
- Verify API calls are actually blocked

. Documentation

- Keep this README updated when adding/removing features
- Document any variant-specific behavior

FAQ

Q: Can I install multiple variants on the same device?
A: No, they share the same `applicationId`. Use debug builds with different IDs for testing.

Q: Which variant should I use for local development?
A: Use `fdroid` or `github` debug builds for full feature access during development.

Q: How do I know which variant is installed?
A: Check Settings → About → Build Info or check version suffix in logs.

Q: Can users switch between variants?
A: No, they must uninstall and reinstall a different variant. Data won't transfer.

Q: Why not use separate apps with different package names?
A: Single package name simplifies distribution across different channels.

Related Documentation

- Build Instructions
- Contributing Guide

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Rhythm Music Player Build System Documentation